

# CareBridge NCG

## Multilingual Cancer-Care Follow-up & Care-Instruction Closure Platform

An assistive, clinician-validated, and human-in-the-loop health delivery platform designed for the National Cancer Grid (NCG) ecosystem. Bridging the critical longitudinal communication gap between complex oncology discharge summaries and patient homes across 11 Indian regional languages.

|                               |                                                                                                         |
|-------------------------------|---------------------------------------------------------------------------------------------------------|
| Author / Developer:           | Jitesh Reddy                                                                                            |
| Target Institution / Context: | National Cancer Grid (NCG) / Tertiary Oncology Centers                                                  |
| Repository:                   | <a href="https://github.com/Jiteshreddy123/CareBridge">https://github.com/Jiteshreddy123/CareBridge</a> |
| Live Web Application:         | <a href="https://jiteshreddy123.github.io/CareBridge/">https://jiteshreddy123.github.io/CareBridge/</a> |
| Core Technology Stack:        | React 19, TypeScript, FastAPI, Python 3.10+, Tailwind CSS                                               |
| Clinical Boundary:            | Assistive, Non-Diagnostic (Mandatory Clinician Sign-off)                                                |
| Submission Type:              | B.Tech Capstone / Clinical AI Software Innovation                                                       |

# 1. Executive Summary

In high-volume public cancer hospitals across India, treatment failure and disease progression often stem not from procedural shortcomings inside hospitals, but from what happens *after* discharge. Oncology discharge summaries are densely filled with English medical jargon, abbreviations (e.g., CBC w/ diff, PET-CT contrast, Cap Capecitabine), and handwritten doctor notes. Over 75% of patients and family caregivers travel from rural or regional districts where Hindi, Telugu, Bengali, Tamil, or Kannada are the primary tongues.

Misunderstanding crucial prerequisite steps—such as fasting for a PET-CT, scheduling pre-chemo blood tests, or recognizing febrile neutropenia—leads to rescheduled appointments, acute toxicities, and alarming treatment dropout rates. **CareBridge NCG** resolves this crisis by transforming hospital orders into structured, doctor-approved, plain-language care passes across 11 Indian languages with closed-loop telemetry and tracking.

**CORE CLINICAL ETHOS:** CareBridge operates strictly within assistive, non-diagnostic boundaries. Every patient card requires treating oncologist verification before dispatch. The platform provides human-in-the-loop safety, ensuring zero unreviewed instructions reach patients.

## 2. Clinical Problem Landscape

- **1. Language & Cognitive Literacy Gap:** High anxiety combined with complex English discharge instructions leads to protocol violations such as eating prior to contrast imaging scans.
- **2. Silent Post-Discharge Dropouts:** Care teams have zero visibility into whether patients completed prerequisite lab work until they arrive on chemotherapy day—frequently causing day-of cancellations.
- **3. Fragmented Caregiver Support:** Family caregivers manage drug schedules and hospital logistics with zero dedicated tooling or delegation mechanisms.
- **4. Unstructured Escalation Channels:** Patients experiencing common chemo side effects rush unnecessarily to emergency rooms or delay seeking help until symptoms become critical.

### 3. System Architecture & Tech Stack

CareBridge NCG is engineered with a modular, decoupled full-stack architecture designed for enterprise reliability and offline-first progressive accessibility on low-tier mobile devices.

| Component              | Technology                               | Responsibility                                                                                              |
|------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Frontend Client        | React 19, TypeScript, Vite, Tailwind CSS | Interactive clinician hub, simulated mobile patient pass, 11-language i18n switcher, audio narration hooks. |
| Backend REST API       | FastAPI, Python 3.10+, Uvicorn, Pydantic | Asynchronous endpoints, schema validation, session tracking, CORS security.                                 |
| Assistive NLP Engine   | Clinical Oncology Lexicon Extractor      | Extracts actions for blood tests, radiation, chemotherapy reviews, and scans with structured explanations.  |
| Audit & Safety Service | Immutable Event Ledger                   | Logs every clinician verification timestamp, user modification, and triage escalation.                      |
| Production Hosting     | GitHub Pages / Cloud CDN                 | Zero-dependency standalone deployment for immediate clinician and stakeholder access.                       |

### 4. Core Functional Modules

#### Module 1: Clinician Governance Portal

Allows oncologists to ingest discharge orders. An assistive NLP parser extracts discrete tasks (CBC, PET-CT, Chemo Day 1). The clinician inspects, fine-tunes, and signs off with a 1-click verification protocol.

#### Module 2: Multilingual Patient Digital Pass

Provides patients with clean, mobile-optimized action cards in 11 Indian languages. Every task specifies: **What** (clear instruction), **Why** (clinical reason), **Where** (department/floor), and **Preparation** (fasting rules). Includes voice narration for low-literacy patients.

#### Module 3: Closed-Loop Tracking Queue

A live command center for hospital care coordinators. Categorizes patient tasks into Pending, Completed, and Overdue. Proactively identifies patients at risk of missing chemotherapy cycles 48 hours in advance.

#### Module 4: Urgent Help & Triage Desk

Enables patients or caregivers to trigger a one-tap 'Need Help' request for symptoms (fever, nausea) or logistics. Triage indicators prioritize high-acuity inquiries for immediate nurse callback.

#### Module 5: Clinical Governance & Audit Trail

An immutable audit log recording exact timestamps, doctor IDs, and verification statuses for every instruction generated, adhering to clinical compliance and healthcare data standards.

## 5. Projected Clinical Outcomes & Impact Metrics

| Metric Indicator                    | Standard Discharge Baseline              | With CareBridge NCG Platform                           |
|-------------------------------------|------------------------------------------|--------------------------------------------------------|
| Treatment Dropout Rate              | 28% - 35% in public registries           | < <b>12%</b> (Projected 60% relative reduction)        |
| Protocol Violations (Fasting/Scans) | 18% - 24% rescheduling rate              | < <b>4%</b> through plain-language What & Why cards    |
| Language Reach                      | Limited to English / basic Hindi         | <b>11 Indian Regional Languages</b> supported          |
| Clinician Counseling Overhead       | 15-20 min repetitive manual explanations | < <b>30 seconds</b> automated draft & 1-click sign-off |
| Adherence Telemetry                 | 0% post-discharge visibility             | <b>100% Closed-Loop</b> tracking queue                 |

## 6. Verification & Demonstration Links

The application has been verified for both local full-stack development and zero-dependency cloud execution:

- **Live Web Demo:** <https://jiteshreddy123.github.io/CareBridge/> (Fully interactive, clinician portal & patient mobile simulation)
- **GitHub Repository:** <https://github.com/Jiteshreddy123/CareBridge> (Full source code, API services, and documentation)
- **FastAPI Swagger Docs:** <http://localhost:8000/docs> (Interactive OpenAPI endpoints for tasks, patients, and audit trails)

## 7. Conclusion

CareBridge NCG demonstrates how thoughtful, human-centric software engineering can overcome systemic healthcare inequities. By respecting clinical boundaries through human-in-the-loop validation while democratizing medical instructions across 11 Indian languages, CareBridge ensures that geographic, economic, and linguistic barriers never stand between a cancer patient and their curative journey.